

SentinelAI Architecture

AI-Powered Behavioral Threat Detection & Investigation Platform

Source code: [GitHub repository](#)

```
MERMAID
flowchart TB
    subgraph Offline["Offline preparation"]
        GEN["Synthetic organization + attacks"] --> EVENTS["events.csv"]
        GEN --> LABELS["labels.csv - evaluation only"]
        EVENTS --> PROFILE["Entity and peer profiles"]
        PROFILE --> FEATURES["Sequential features"]
        FEATURES --> IF["Isolation Forest"]
        FEATURES --> RF["Class-weighted Random Forest"]
        LABELS --> EVAL["Evaluation"]
        IF --> EVAL
        RF --> EVAL
    end

    subgraph Online["Near-real-time SOC path"]
        REPLAY["Sequential event replay"] --> FE["Feature + rolling history"]
        FE --> UNKNOWN["Isolation Forest"]
        FE --> RULES["Deterministic attack rules"]
        FE --> CLASSIFY["Attack classifier"]
        UNKNOWN --> RISK["0-100 explainable risk"]
        RULES --> RISK
        CLASSIFY --> RISK
        RISK --> SQLITE["SQLite events, alerts, replay state"]
        SQLITE --> API["FastAPI /api/v1"]
        API --> DASH["Streamlit analyst dashboard"]
    end
```

Cold-start resolution follows entity -> department -> entity type -> organization. Trusted normal events update recent baselines through exponential decay; anomalous events are excluded from profile updates.